

Neural Text

Category : ARCHITECTURE

Points : 100

Q1: This layer type maps discrete tokens like words or characters to dense, continuous vectors whose dimensions are learned during training, encoding, e.g., semantic similarity.

A: What is an embedding layer.

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Q2: This is the key architectural feature that distinguishes RNNs from traditional feedforward networks, allowing them to process variable sequences of data.

A: What is a feedback loop or recurrent / recursive connection.

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Q3: This RNN architecture processes input sequences both forward and backward, helping capture context from both past and future tokens in tasks like text classification.

A: What is a Bidirectional RNN.

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Q4: This RNN variant added gating mechanisms and a residual connection to address the vanishing gradient problem in long sequences.

A: What is LSTM.

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Q5: This simpler cousin of the LSTM uses fewer parameters by combining the forget and input gates into a single update gate.

A: What is GRU (gated recurrent unit).

Category : EMBEDDINGS

Points : 100

Q6: True or False: one-hot encodings of text are 'sparse' representations.

A: True.

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Q7: True or False: One way to obtain image embeddings is by passing images through the convolutional base layers of a pre-trained CNN.

A: True.

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Q8: With this text-embedding approach, one uses surrounding words (the context) to predict a target word, or uses a word to predict the surrounding words (the context).

A: What is Word2Vec

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Q9: When instantiating an RNN layer in Keras, e.g., `keras.layers.LSTM(8)`, this is what the first positional argument represents (the 8 in this case), which amounts to a form of sequence embedding.

A: What is the dimensionality (length) of the hidden state vector?

Category : EMBEDDINGS

Points : 500

Q10: The weights in an embedding layer are refined/learned through this process.

A: What is back propagation?

Category : APPLICATIONS

Points : 100

Q11: Google Translate implemented this type of RNN in 2016, replacing a prior rule-based engine.

A: What is an LSTM?

Category : APPLICATIONS

Points : 200

Q12: RNNs are especially useful for tasks like language modeling or stock prediction because they can capture this relationship between inputs.

A: What is temporal or sequential information.

Category : APPLICATIONS

Points : 300

Q13: True or False: RNNs can be used to model audio data.

A: True.

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Points : 400

Q14: Early transformers were inferior to RNNs for real-time translation or speech tasks primarily because of this performance issue.

A: What is high latency.

Category : APPLICATIONS

Points : 500

Q15: The first use of attention in neural networks was applied to this architecture.

A: What is an RNN! Bahdanau et al. (2014). Neural Machine Translation by Jointly Learning to Align and Translate.